Project: Skylar Streaming Telemetry

Vision

Observability gives way to the observer. Skylar Advisor is the observer. – Larry Lancaster

Customers using Skylar Advisor will no longer need to wade through mountains of data to prevent and fix functional and performance issues in their managed entities. Skylar will gather data, generate insights, and provide information and resolutions to customers without needing to be glued to a computer screen all day. To reduce noise for customers Skylar needs entity data and contextual data that allows it to understand identity, state, and relationships of those entities. Streaming telemetry is increasingly becoming a useful source for this type of data, but SL1 is not equipped to handle it. The goal of this project is to take advantage of the many streaming data sources that can help drive up the value of Skylar and integrate them with SL1 where it makes sense.

Approach

Skylar has laid the ground floor for supporting streaming telemetry, and this project will complete the next levels. We will allow customers to send telemetry about SL1 managed devices and about entities that SL1 cannot easily monitor. We will use the high-granularity, contextually-rich telemetry to drive better Skylar Advisor insights similar to Skylar Log RCA. As streaming telemetry becomes more prevalent and ScienceLogic becomes a competitor in this area of observability, we will continue to add features that support more integrations with SL1, with Skylar Advisor, and other third-party products.

Business Strategy

Our streaming telemetry strategy will be similar to Zebrium's approach to log analytics: focus on the value that curated insights can provide rather than data exploration and management. The market is saturated with solutions that store, analyze, and visualize telemetry, but sparse on solutions that provide automated root cause analysis for de-noising daily operations. Anomaly detection, predictive alerts, custom alerts, and reports will all be more efficient and powerful if

they can use new rich datasources like OpenTelemetry, Prometheus, Osquery, and eBPF in conjunction with existing collections from SL1.

Phases

We will take a phased approach that allows us to get comfortable with the business space, technology, and customer use cases while we build out a long-term roadmap.

Phase 1: Basic Host Metrics

Phase details: Streaming Telemetry Phase 1

Work items: SKY-1199: Streaming Telemetry Phase 1: Basic Data and Analytics DONE

Deliverables:

- Skylar Collector (agent) binary with OpenTelemetry Collector and Osquery
- OTELCOL with config for linux and Windows collecting host metrics
- OTEL analytics pipeline for generic OTEL entities and time series identification
- Host dataset with cpu, memory, disk utilization

Phase 2: Insight Generation

In this phase we will use the basic OTEL pipeline to generate insights for Skylar Advisor and integrated with SL1.

Deliverables:

- Skylar Advisor insights:
  - Anomaly detection
  - Predictive alerts
  - Threshold alerts
- Automatic dataset generation
- Osquery included in datasets
- Basic event management
- SL1 event integration

Phase 3: Relationships Insights

We expand insights into topology and entity relationships.

Deliverables:

- Skylar Advisor insights:
  - Topology insights

  - Relationship insights
- eBPF support in Skylar Collector

Problem Spaces

These are the challenges that we need to solve in order to effectively use streaming telemetry in ScienceLogic products. Our solution should address them.

Entity Identification and Classification

We own the data from source to destination with polling collection, making it easy to identify the source entity and know how to derive insights from it. The source data and the processing destination are loosely coupled in the streaming telemetry paradigm, therefore we must take advantage of technology and architecture to provide identity and classification that drives insights.

Agent Onboarding

We need to associate new streams of data into Skylar with the corresponding SL1 stack. This means that somewhere in the software stack, we need to insert an ID in the telemetry that easily maps to a customer and stack.

In addition to the ID, we need to decide if we will manage the remote agents or if that is a customer concern. While this is a product design decision, there are a number of technologies we can use for this purpose so we will need to explore them.

Data Sources

OpenTelemetry is a vendor-neutral set of specs, formats, and software that provides instrumentation, data transport, routing, and guidelines for how to programmatically process telemetry. This is an obvious choice for us to support.

What about other formats? Prometheus, osquery, etc.?

Thresholds and Alerting

While we have many examples of how other products handle thresholds and alerts, including SL1, we have an opportunity to rethink our approach to implementing them. See Streaming Telemetry: Alerting.

Integrations

SL1 will be our primary integration to start, but we will eventually support third-party telemetry and API-based applications. We will need to provide access to the data and insights.

Design

The high-level design follows Skylar's design. Streaming Telemetry will feed into the Skylar analytics pipeline and feed anomaly detection, predictive alerting, and ultimately Skylar Advisor similarly to SL1 data. Handling streaming telemetry from sources other than SL1 will require a new entity identification process and some changes to the OLAP process to incorporate this new generalized data.

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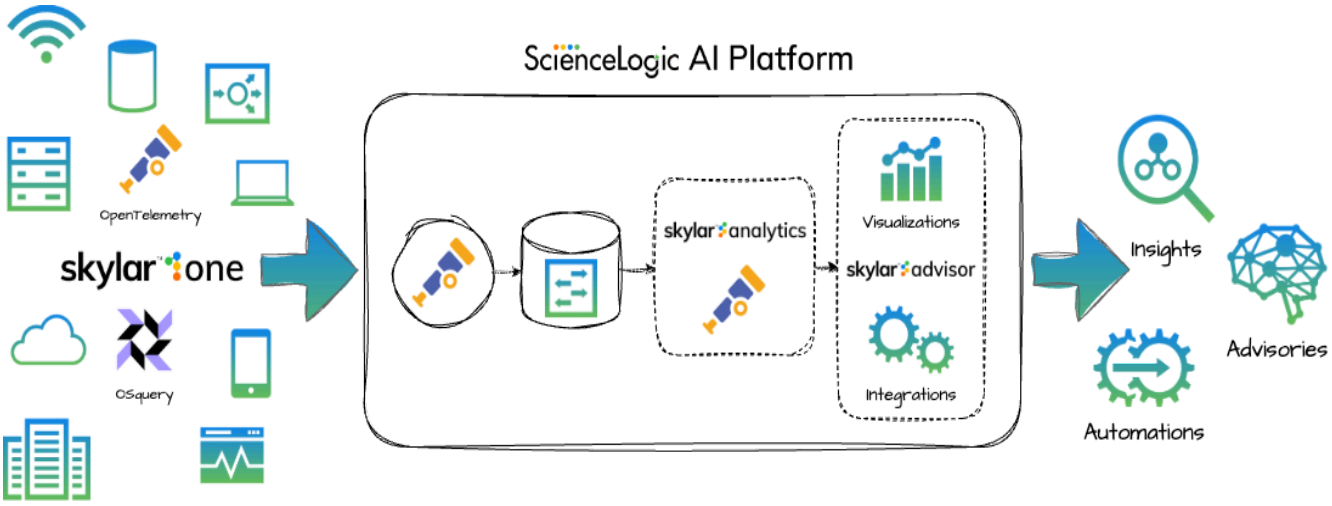

Important Links

For the product input, see SL1/Skylar Glue for Streaming Telemetry and Streaming Telemetry Product Definition - MVP.

For project decisions, see ADRs for Skylar Streaming Telemetry.

Child Pages

Analytics Notes

Host Fingerprint Testing

Infrastructure Observability Receivers

Natural Language Query and Dashboard Generation (innovation project)

Nexus Live 2025 - Streaming Telemetry

OpenTelemetry and OLAP Cubes

Osquery Sources

Skylar OTEL Collector

SL1 OTEL Analytics Pipeline Walkthrough

Snippet Framework Collector

Streaming Logs

Streaming Telemetry: Alerting

Streaming Telemetry Phase 1

Streaming Telemetry Product Definition - MVP

Streaming Telemetry: Release Planning